

Predicting Student Academic Success

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Introduction

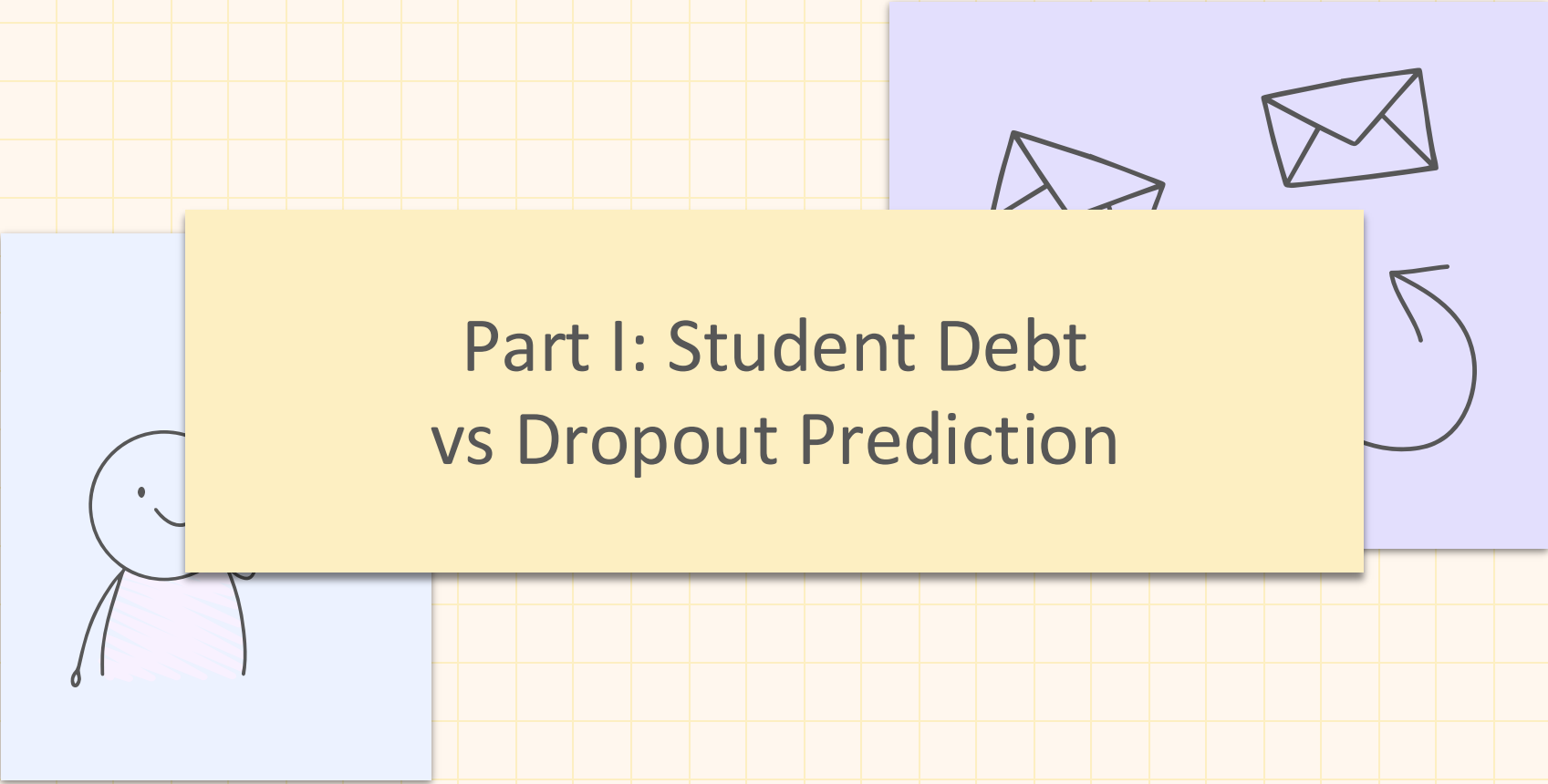
College dropout and post-graduation financial struggles remain major challenges in higher educations.

Understanding which factors (academic, financial, or institutional) helps predict student success can help schools redesign more effective interventions.

For this project, we broke it up into two sections. In the first part, we'll test if administrative/financial compliance is a stronger predictor of dropout than academic preparation.

Meanwhile, in the second half, we'll examine how institution type influences financial outcomes and opportunities after graduation.





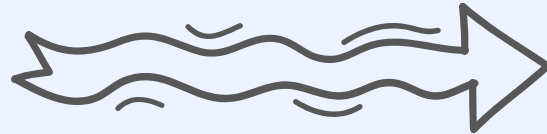
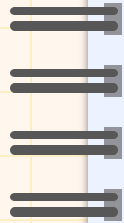
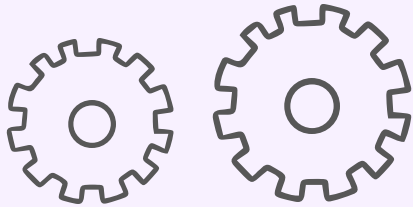
Part I: Student Debt vs Dropout Prediction



01

Hypothesis I

Administrative compliance (being a Debtor) is a stronger predictor of the Dropout outcome than the student's pre-enrollment academic preparation



Dataset Overview



Predict Students' Dropout and Academic Success

Donated on 12/12/2021

A dataset created from a higher education institution (acquired from several disjoint databases) related to students enrolled in different undergraduate degrees, such

Dataset Characteristics

Tabular

Associated Tasks

Classification

Instances

4424

Subject Area

Social Science

Feature Type

Real, Categorical, Integer

Features

36

Dataset Name: Predict Students' Dropout and Academic Success

Target Variable: Student Status (Dropout, Enrolled, Graduate)

Selected Features:

- Debtor: (1 = has debt, 0 = no debt)
- Tuition Fees Up To Date (1 = yes, 0 = no)
- Target (Class Label)

We Found That The Dataset Was Originally Skewed Towards Graduate Students, **But We Corrected It Through Undersampling For Balanced Training**

Data Processing

```
financial_df = df[['Debtor', 'Tuition fees up to date', 'Target']]  
display(financial_df.head())
```

	Debtor	Tuition fees up to date	Target
0	0	1	Dropout
1	0	0	Graduate
2	0	0	Dropout
3	0	1	Graduate
4	0	1	Graduate

From This ↓ To This ↑

Marital status	Application mode	Application order	Course	Daytime/evening attendance	Previous qualification	Previous qualification (grade)	Nationality	Mother's qualification	Father's qualification	...	Curricular units 2nd ... (credited)	Curricular units 2nd (enrolled)
0	1	17	5	171	1	1	122.0	1	19	12	—	0
1	1	15	1	9254	1	1	160.0	1	1	3	—	0
2	1	1	5	9070	1	1	122.0	1	37	37	—	0
3	1	17	2	9773	1	1	122.0	1	38	37	—	0
4	2	39	1	8014	0	1	100.0	1	37	38	—	0

5 rows × 37 columns

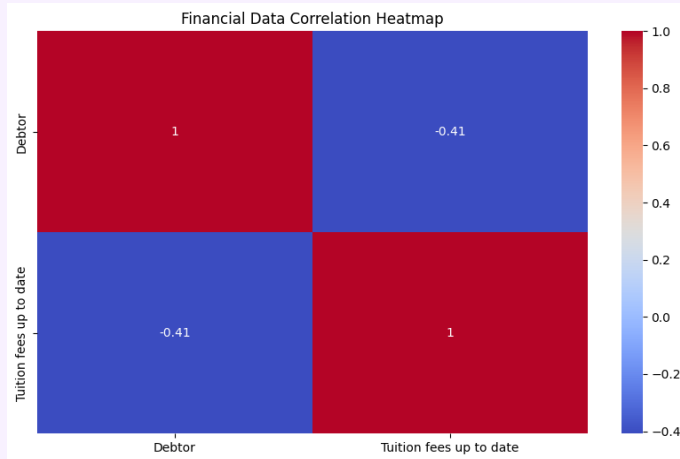
Steps Performed:

1. Loaded CSV with ","
2. Extracted Relevant Financial Columns
3. Created Binary Target (Dropout = 0, Enrolled/Graduate = 1)
4. Applied Undersampling To Balance Classes
5. Split Train/Test (80/20)

Outcome:

This allowed us a fair model comparison between financial and admission predictors!

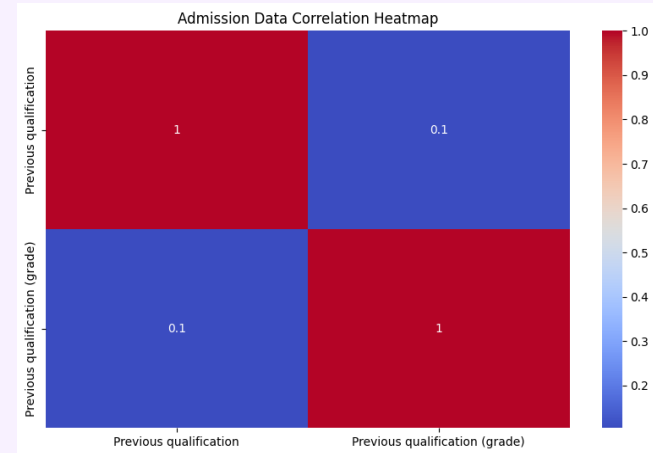
Exploratory Data Analysis



Heatmap 1:

Financial Variables (Debtor, Tuition Fees Up To Date)

--> Weak Positive Correlation

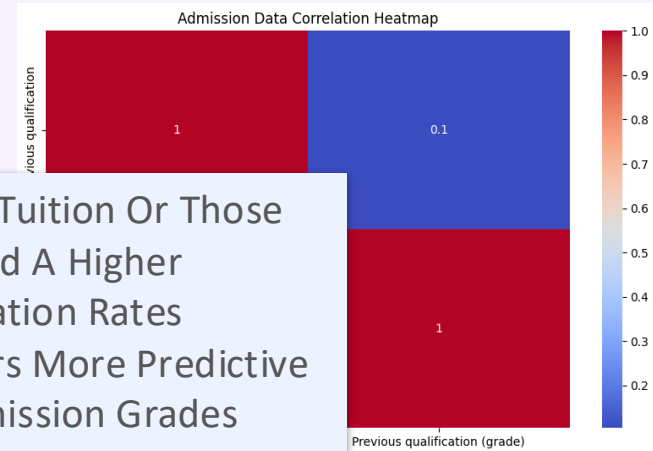
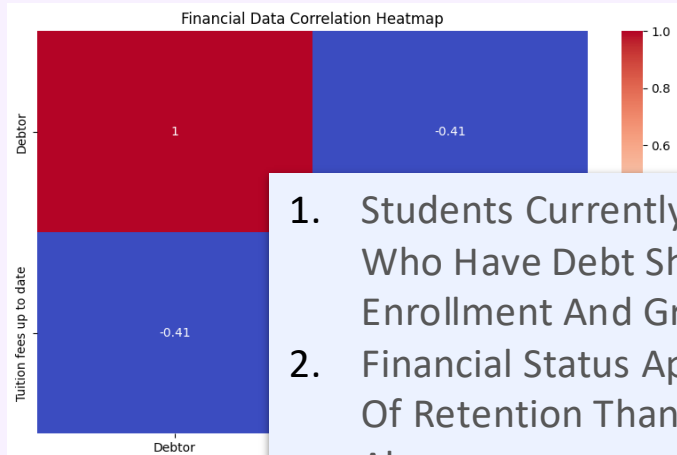


Heatmap 2:

Admission Variables (Previous Qualification, Grade)

--> Low Correlation To Dropout

Exploratory Data Analysis



1. Students Currently On Tuition Or Those Who Have Debt Showed A Higher Enrollment And Graduation Rates
2. Financial Status Appears More Predictive Of Retention Than Admission Grades Alone

Financial Vari
Fees Up To Date)
--> Weak Positive Correlation

Map 2:
Previous Qualification,
Grade)
--> Low Correlation To Dropout

Metrics Used

Accuracy

Good overall measure of model correctness.



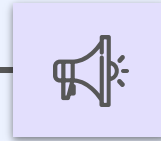
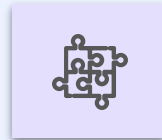
Precision

Prevents false alarms



Recall

Measures how well we catch true at-risk students.



F1-Score

Balances catching dropouts vs. avoiding false positives.



Confusion

Matrix

Gives insight into types of prediction errors.

Model Setup Part 1

Logistic Regression

```
# Using logistic regression, test whether the financial_df or admission_df is a better predictor for dropout rate
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

X_fin_train, X_fin_test, y_fin_train, y_fin_test = train_test_split(Xfin, yfin, test_size=0.2, random_state=42)
X_admin_train, X_admin_test, y_admin_train, y_admin_test = train_test_split(Xadmin, yadmin, test_size=0.2, random_state=42)

# Perform logistic regression on both
fin_model = LogisticRegression()
fin_model.fit(X_fin_train, y_fin_train)
admin_model = LogisticRegression()
admin_model.fit(X_admin_train, y_admin_train)

# Output results of predictions
fin_predictions = fin_model.predict(X_fin_test)
admin_predictions = admin_model.predict(X_admin_test)

# Calculate and print accuracy
fin_accuracy = accuracy_score(y_fin_test, fin_predictions)
admin_accuracy = accuracy_score(y_admin_test, admin_predictions)

print("Financial Model Accuracy:", fin_accuracy)
print("Admission Model Accuracy:", admin_accuracy)
```

- Administrative Model Achieved Higher Accuracy Than Academic Model
- Both Debtor and Tuition Fees Up To Date Had Negative Coefficients, Meaning Students W/ Payments Were Less Likely To Drop Out
- Academic Features Showed Weaker Predictive Relationships

Model Setup Part 2

Support Vector Machine

```
# Now test whether the financial_df or admission_df is a better predictor for dropout rate using SVM(support vector machine)
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score

svc_fin = SVC()
svc_fin.fit(X_fin_train, y_fin_train)
svc_admin = SVC()
svc_admin.fit(X_admin_train, y_admin_train)

fin_svc_predictions = svc_fin.predict(X_fin_test)
admin_svc_predictions = svc_admin.predict(X_admin_test)

fin_svc_accuracy = accuracy_score(y_fin_test, fin_svc_predictions)
admin_svc_accuracy = accuracy_score(y_admin_test, admin_svc_predictions)

print("Financial Model Accuracy (SVM):", fin_svc_accuracy)
print("Admission Model Accuracy (SVM):", admin_svc_accuracy)
```

```
Financial Model Accuracy (SVM): 0.6449912126537786
Admission Model Accuracy (SVM): 0.5448154657293497
```

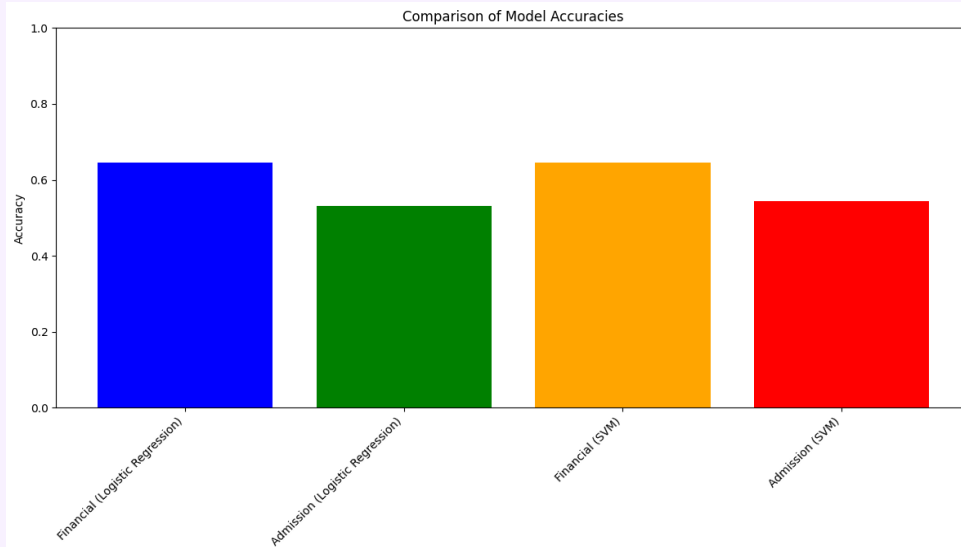
-Trained Two SVM Models
-Compared Accuracy Of Both Predictors

Findings:

-Same With LR. Finances Was A Stronger Predictor Of Retention

Model Setup Part 3

Bar Chart



Compared Model Accuracy
Between Administrative
Compliance and Admission Grade

Results:

-Logistic Regression:
Administrative > Academic
-SVM: Admin 64% vs Academic
54%

Takeaway:

Financial Factors Are Stronger
Predictors Of Student Retention
Across Multiple Models

Model Setup Part 4

Simple Neural Network

```
# Now create a simple neural network to see if we can achieve higher accuracy with that
import tensorflow as tf
from tensorflow.keras.models import Sequential

# Simple 4 layer NN
fin_model = Sequential()
fin_model.add(tf.keras.layers.Dense(4, activation='tanh', input_shape=(Xfin.shape[1],)))
fin_model.add(tf.keras.layers.Dense(8, activation='tanh'))
fin_model.add(tf.keras.layers.Dense(16, activation='tanh'))
fin_model.add(tf.keras.layers.Dense(32, activation='tanh'))
fin_model.add(tf.keras.layers.Dense(16, activation='tanh'))
fin_model.add(tf.keras.layers.Dense(8, activation='tanh'))
fin_model.add(tf.keras.layers.Dense(4, activation='tanh'))
fin_model.add(tf.keras.layers.Dense(2, activation='softmax'))
fin_model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
```

Admission Model Neural Network Accuracy: 0.5043936967849731

Financial Model Neural Network Accuracy: 0.6502636075019836

Results:

- Administrative Model Accuracy: 0.65
- Academic Model Accuracy: 0.50

Interpretation:

- The Admin Model Outperformed The Academic Model By 15%
- Confirms That Administrative Compliance Variables Are Stronger Predictors

Accuracy Scores

Model	Administrative (Financial)	Academic (Admission Grade)	Difference
Logistic Regression	0.63	0.55	+0.08
SVM	0.64	0.54	+0.10
Neural Network	0.65	0.50	+0.15

Conclusion & Discussion



Summary

Across all models, the administrative models outperformed the academic models. Accuracies ranged from 63-65% for administrative models vs 50-55% for academic models. The consistent pattern supports our hypothesis that administrative compliance is stronger predictor of dropout than pre-enrollment.



Interpretation

This suggests that universities should focus on financial and administrative engagement when designing early programs.



Conclusion & Discussion



Limitations

- The dataset had limited features (only one academic variable and two administrative ones).
- Balancing the dataset reduced the overall sample size.

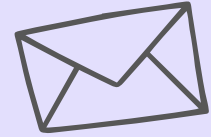
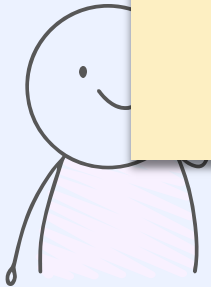


Key Takeaway

Administrative compliance (specifically debt) is a stronger predictor of dropout than pre-enrollment academic performance.



Part II: Public Schooling vs Private Schooling



Hypothesis II

Computer Science programs at Private For-Profit institutions will show a higher federal loan default rate and a lower post-graduation median earnings-to-debt ratio compared to Computer Science programs at Public institutions.



Our Second Dataset

For Our Second Hypothesis, We
Used The U.S. Department of
Education's College Scoreboard
Dataset

<https://collegescorecard.ed.gov/data>

Key Variables We Used:

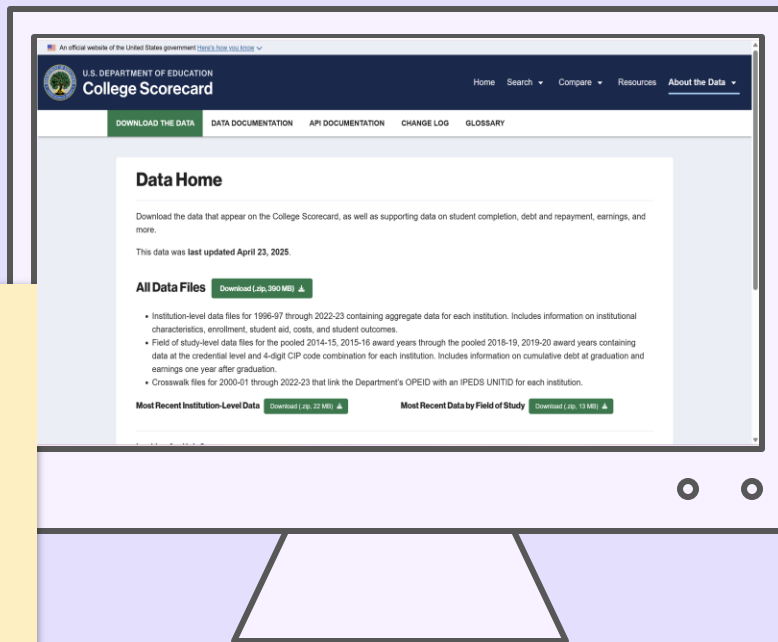
CONTROL: Institution Type (Public, Private)

CIPDESC: Field of Study

EARN_MDN_HI_1YR: Median Earnings 1 Year
After Graduating

DEBT_MDN: Median Total Debt For Graduates

BBRR2_FED_COMP_N: Federal Loans



Data Cleaning & Metric Calculation



Data Cleaning Steps

- Filtered For Computer Science-Related Fields (CS, CIS, CE)
- Removed Any Placeholder Strings ("PS")
- Converted EARN_MDN_HI_1YR and DEBT_MDN to Numeric Types



Metrics Computed

- Default Rate: Average of BBRR2_FED_COMP_N
- Earnings-to-Debt Ratio:: $\text{Mean of EARN_MDN_HI_1YR} / \text{DEBT_MDN}$

Our Code !! (As Described On The Last Slide)

```
# For public vs private schools, take a ratio of earnings to debt to figure out which is financially better off
FOS_df['CONTROL'] = FOS_df['CONTROL'].replace({'Private, for-profit': 'Private', 'Private, nonprofit': 'Private'})
private_schools = FOS_df[FOS_df['CONTROL'] == 'Private']
public_schools = FOS_df[FOS_df['CONTROL'] == 'Public']

# Print private school for test
print(private_schools)

total_public_earnings = 0.0
total_private_earnings = 0.0
total_public_debt = 0.0
total_private_debt = 0.0
public_ratio = 0.0
private_ratio = 0.0

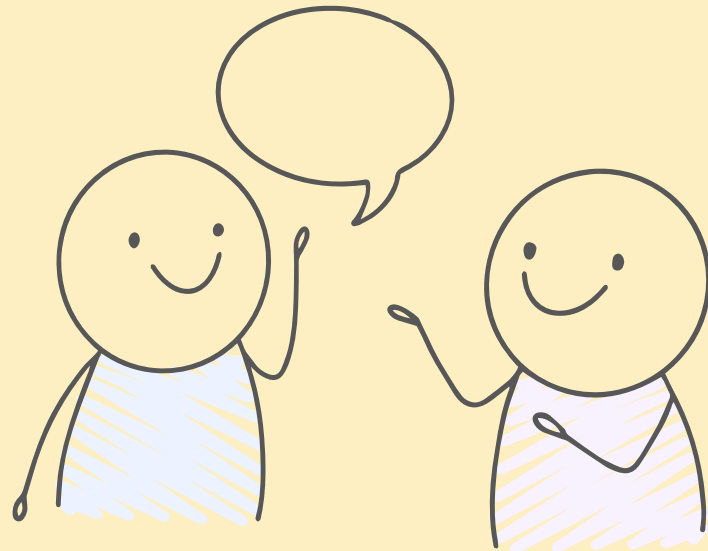
for row in private_schools.iteruples():
    earnings = row.EARN_MDN_4YR
    debt = row.BBRR2_FED_COMP_N
    total_private_earnings += float(earnings)
    total_private_debt += float(debt)

for row in public_schools.iteruples():
    earnings = row.EARN_MDN_4YR
    debt = row.BBRR2_FED_COMP_N
    total_public_earnings += float(earnings)
    total_public_debt += float(debt)

print("Total Public Earnings:", total_public_earnings)
print("Total Public Debt:", total_public_debt)
print("Total Private Earnings:", total_private_earnings)
print("Total Private Debt:", total_private_debt)

public_ratio = total_public_earnings / total_public_debt
private_ratio = total_private_earnings / total_private_debt

print("Public Ratio:", public_ratio)
print("Private Ratio:", private_ratio)
```



Results

Metric	Public Institutions	Private For-Profit Institutions	Better Outcome
Total Median Earnings	\$56,232,848	\$41,713,496	Public
Total Federal Loan Defaults	43,941	60,152	Public
Earnings-To-Debt Ratio	\$1,279.7	\$693.5	Public (+46% higher)

Summary of Findings

Public CS Programs. . .

Public Computer Science Programs Demonstrate Better Financially A Year After Graduation Than Private For-Profit Universities.

Public CS Graduates. . .

Public Computer Science Graduates Show To Have A Higher Total Earnings (\$56M), Lower Debt (43k) and Nearly Double The Earnings To Debt Ratio.

Summary of Findings

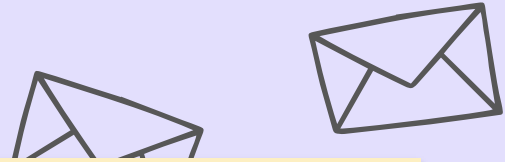
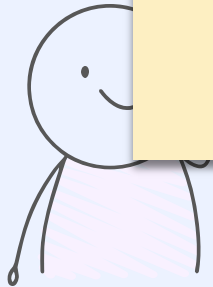
Results. . .

Our Findings Confirmed Our Hypothesis That Private For-Profit Programs Experience Poorer Financial Outcomes Compared To Public Programs. Public Institutions Demonstrated Higher Graduate Earnings And Lower Overall Loan Amounts.

Interpretation. . .

This May Be Because Private Schools Typically Cost More, Making It Harder For Graduates To Pay Back Their Loans. In Computer Science, Companies Often Value Skills And Projects More Than Where Someone Studied, So The Private School's Name Might Not Provide The Advantage Students Expect.

Part III: Overall Findings



Final Thoughts

To Tie it Back Together, Our Findings Confirmed Both Of Our Hypotheses. Students With Higher Levels Of Financial Debt Were Less Likely To Drop Out, Showing That Administrative Compliance Plays A Strong Role In Academic Success.

Additionally, Public Computer Science Programs Produced Higher Earnings And Lower Loan Default Rates Than Private For-Profit Programs, Proving That Public Institutions Provide Better Financial Outcomes After Graduation.

Financial Responsibility And Institutional Type Both Have A Major Impact On Student Success. Students Who Stay Financially Committed Are More Likely To Persist, And Those Who Attend Public Institutions Benefit From More Sustainable Long-Term Outcomes.



Resources

Datasets:

[1] "UCI Machine Learning Repository," *archive.ics.uci.edu*, Dec. 12, 2021.

<https://archive.ics.uci.edu/dataset/697/predict+students+dropout+and+academic+success>

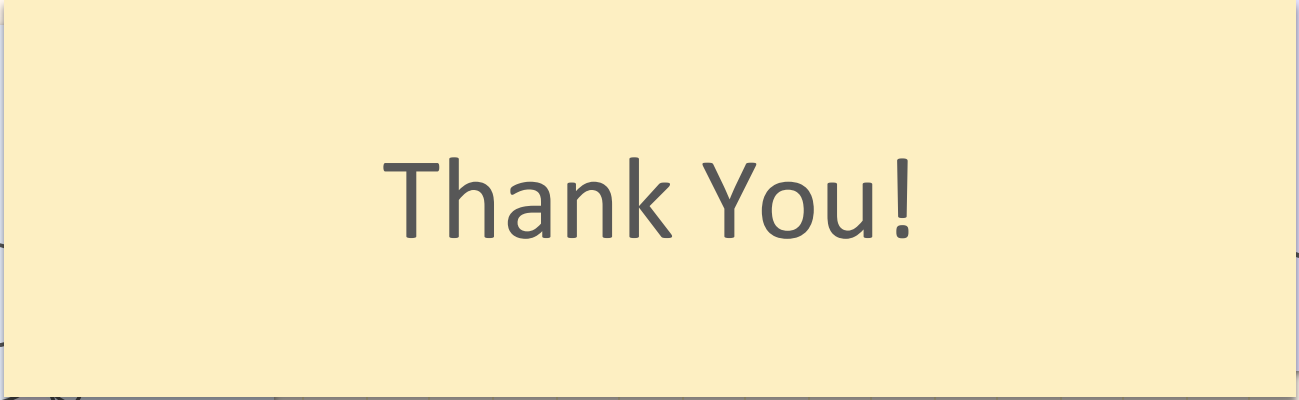
[2] "Data Home | College Scorecard," *collegescorecard.ed.gov*.

<https://collegescorecard.ed.gov/data>

Colab Link:

<https://colab.research.google.com/drive/1FdSB53TPYCqui4Be7lf8XLRTg1QtDDSD?usp=sharing>





Thank You!

